

Electronic Circuits

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1

FETs vs. BJTs

Similarities:

- Amplifiers
- Switching devices
- Impedance matching circuits

Differences:

- FETs are voltage controlled devices.
- BJTs are current controlled devices.
- FETs have a higher input impedance.
- BJTs have higher gains.
- FETs are less sensitive to temperature variations and are more easily integrated on ICs.
- FETs are generally more static sensitive than BJTs.

2

FET Types

• **JFET:** Junction FET

• **MOSFET:** Metal-Oxide-Semiconductor FET

• **D-MOSFET:** Depletion MOSFET

• **E-MOSFET:** Enhancement MOSFET

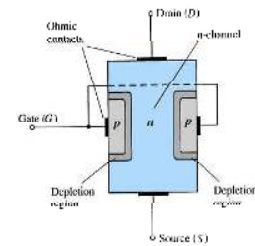
3

JFET Construction

There are two types of JFETs

- **n-channel**
- **p-channel**

The n-channel is more widely used.



There are three terminals:

- **Drain (D)** and **Source (S)** are connected to the n-channel
- **Gate (G)** is connected to the p-type material

4

JFET Operation: The Basic Idea

JFET operation can be compared to a water spigot.

The source of water pressure is the accumulation of electrons at the negative pole of the drain-source voltage.

The drain of water is the electron deficiency (or holes) at the positive pole of the applied voltage.

The control of flow of water is the gate voltage that controls the width of the n-channel and, therefore, the flow of charges from source to drain.



5

JFET Operating Characteristics

There are three basic operating conditions for a JFET:

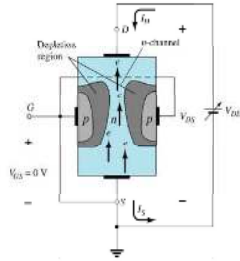
- $V_{GS} = 0$, V_{DS} increasing to some positive value
- $V_{GS} < 0$, V_{DS} at some positive value
- Voltage-controlled resistor

6

JFET Operating Characteristics: $V_{GS} = 0\text{ V}$

Three things happen when $V_{GS} = 0$ and V_{DS} is increased from 0 to a more positive voltage

- The depletion region between p-gate and n-channel increases as electrons from n-channel combine with holes from p-gate.
- Increasing the depletion region, decreases the size of the n-channel which increases the resistance of the n-channel.
- Even though the n-channel resistance is increasing, the current (I_D) from source to drain through the n-channel is increasing. This is because V_{DS} is increasing.

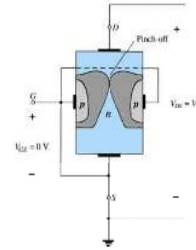


7

JFET Operating Characteristics: Pinch Off

If $V_{GS} = 0$ and V_{DS} is further increased to a more positive voltage, then the depletion zone gets so large that it **pinches off** the n-channel.

This suggests that the current in the n-channel (I_D) would drop to 0A, but it does just the opposite—as V_{DS} increases, so does I_D .

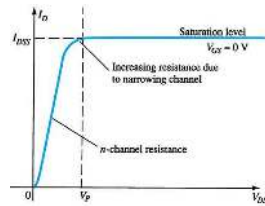


8

JFET Operating Characteristics: Saturation

At the pinch-off point:

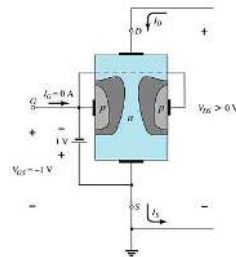
- Any further increase in V_{GS} does not produce any increase in I_D . V_{GS} at pinch-off is denoted as V_P .
- I_D is at saturation or maximum. It is referred to as I_{DSS} .
- The ohmic value of the channel is maximum.



9

JFET Operating Characteristics

As V_{GS} becomes more negative, the depletion region increases.

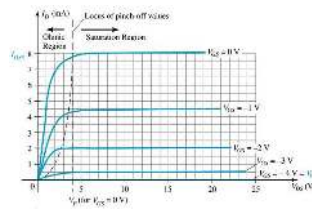


10

JFET Operating Characteristics

As V_{GS} becomes more negative:

- The JFET experiences pinch-off at a lower voltage (V_P).
- I_D decreases ($I_D < I_{DSS}$) even though V_{DS} is increased.
- Eventually I_D reaches 0 A. V_{GS} at this point is called V_P or $V_{GS(off)}$.



Also note that at high levels of V_{DS} the JFET reaches a breakdown situation. I_D increases uncontrollably if $V_{DS} > V_{DSmax}$.

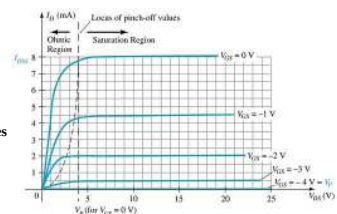
11

JFET Operating Characteristics: Voltage-Controlled Resistor

The region to the left of the pinch-off point is called the **ohmic region**.

The JFET can be used as a variable resistor, where V_{GS} controls the drain-source resistance (r_d). As V_{GS} becomes more negative, the resistance (r_d) increases.

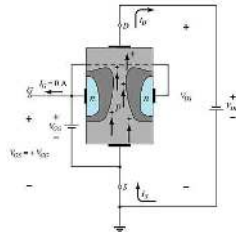
$$r_d = \frac{r_0}{\left(1 - \frac{V_{GS}}{V_P}\right)^2}$$



12

p-Channel JFETS

The p-channel JFET behaves the same as the n-channel JFET, except the voltage polarities and current directions are reversed.

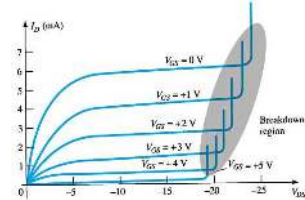


13

p-Channel JFET Characteristics

As V_{GS} increases more positively

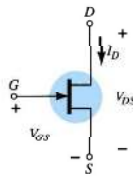
- The depletion zone increases
- I_D decreases ($I_D < I_{DSS}$)
- Eventually $I_D = 0$ A



Also note that at high levels of V_{DS} the JFET reaches a breakdown situation: I_D increases uncontrollably if $V_{DS} > V_{DSmax}$.

14

N-Channel JFET Symbol



15

JFET Transfer Characteristics

The transfer characteristic of input-to-output is not as straightforward in a JFET as it is in a BJT.

In a BJT, β indicates the relationship between I_B (input) and I_C (output).

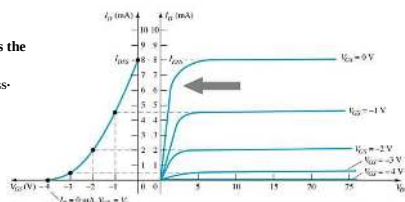
In a JFET, the relationship of V_{GS} (input) and I_D (output) is a little more complicated:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2$$

16

JFET Transfer Curve

This graph shows the value of I_D for a given value of V_{GS} .



17

Plotting the JFET Transfer Curve

Using I_{DSS} and V_p ($V_{GS(off)}$) values found in a specification sheet, the transfer curve can be plotted according to these three steps:

Step 1

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2$$

Solving for $V_{GS} = 0$ $I_D = I_{DSS}$

Step 2

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2$$

Solving for $V_{GS} = V_p$ ($V_{GS(off)}$) $I_D = 0$ A

Step 3

$$\text{Solving for } V_{GS} = 0 \text{ V to } V_p \quad I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2$$

18

JFET Specifications Sheet

Electrical Characteristics

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted)					
Characteristic	Symbol	Min	Typ	Max	Unit
OFF CHARACTERISTICS					
Gate-Source Breakdown Voltage ($I_G = 0, V_{DS} = 0$)	$V_{GS(off)}$	-25	—	—	Vdc
Gate-Source Current ($V_{GS} = -15\text{ Vdc}, V_{DS} = 0$ ($V_{GS} = -15\text{ Vdc}, V_{DS} = 0, T_A = 100^\circ\text{C}$)	I_{GSS}	—	—	-1.0	μAdc
Gate-Source Cutoff Voltage ($V_{DS} = 15\text{ Vdc}, I_G = 10\text{ nAdc}$)	$V_{GS(off)}$	-0.5	—	-6.0	Vdc
Gate-Source Voltage ($V_{GS} = 15\text{ Vdc}, I_G = 10\text{ } \mu\text{Adc}$)	V_{GS}	—	-2.5	—	Vdc
ON CHARACTERISTICS					
On-State Voltage Drop Coefficient ($V_{GS} = 15\text{ Vdc}, V_{DS} = 0$)	$V_{DS(on)}$	1.0	1.0	2.0	mVdc
SMALL-SIGNAL CHARACTERISTICS					
Forward Transfer Admittance Coefficient ($V_{GS} = 15\text{ Vdc}, V_{DS} = 0, f = 1\text{ kHz}$)	Y_{fs}	1000	—	2000	μS
Output Admittance Coefficient ($V_{GS} = 15\text{ Vdc}, V_{DS} = 0, f = 1\text{ kHz}$)	Y_{os}	—	10	20	μS
Input Capacitance ($V_{GS} = 15\text{ Vdc}, V_{DS} = 0, f = 1\text{ kHz}$)	C_{iss}	—	4.5	10	pF
Reverse Transfer Capacitance ($V_{GS} = 15\text{ Vdc}, V_{DS} = 0, f = 1\text{ kHz}$)	C_{rss}	—	1.5	2.0	pF

(Values are typical for 2N5457 and 2N5458)

19

JFET Specifications Sheet

Maximum Ratings

Rating	Symbol	Value	Unit
Drain-Source Voltage	V_{DS}	25	Vdc
Drain-Gate Voltage	V_{DG}	25	Vdc
Reverse Gate-Source Voltage	V_{GS}	-25	Vdc
Gate Current	I_G	10	mAdc
Total Device Dissipation @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	310	mW
Junction Temperature Range	T_J	125	$^\circ\text{C}$
Storage Channel Temperature Range	T_{stg}	-55 to +150	$^\circ\text{C}$

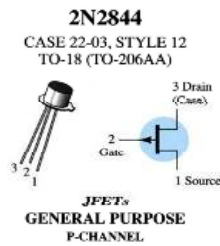


Refer to 2N4220 for graphs.

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20

Case and Terminal Identification



21

Testing JFETs

- Curve Tracer**
A curve tracer displays the I_D versus V_{DS} graph for various levels of V_{GS} .
- Specialized FET Testers**
These testers show I_{DSS} for the JFET under test.

22

MOSFETs

MOSFETs have characteristics similar to JFETs and additional characteristics that make them very useful.

There are two types of MOSFETs:

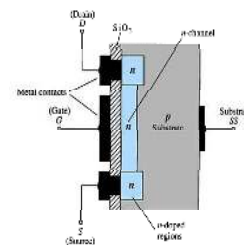
- Depletion-Type
- Enhancement-Type

23

Depletion-Type MOSFET Construction

The **Drain (D)** and **Source (S)** connect to the n -doped regions. These n -doped regions are connected via an n -channel. This n -channel is connected to the **Gate (G)** via a thin insulating layer of SiO_2 .

The n -doped material lies on a p -doped substrate that may have an additional terminal connection called **Substrate (SS)**.

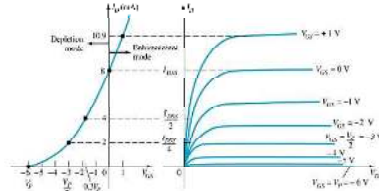


24

Basic MOSFET Operation

A depletion-type MOSFET can operate in two modes:

- Depletion mode
- Enhancement mode

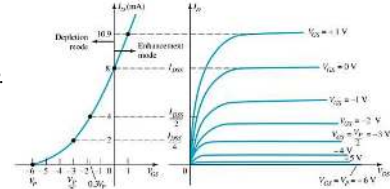


25

D-Type MOSFET in Depletion Mode

Depletion Mode

The characteristics are similar to a JFET.



- When $V_{GS} = 0$ V, $I_D = I_{DSS}$
- When $V_{GS} < 0$ V, $I_D < I_{DSS}$
- The formula used to plot the transfer curve still applies:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$

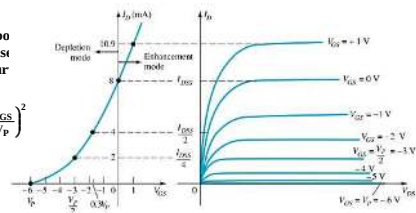
26

D-Type MOSFET in Enhancement Mode

Enhancement Mode

- $V_{GS} > 0$ V
- I_D increases as V_{GS} increases
- The formula used to plot the transfer curve still applies:

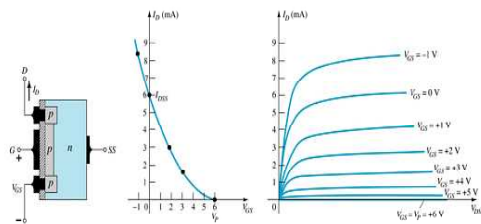
$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2$$



Note that V_{GS} is now a positive polarity

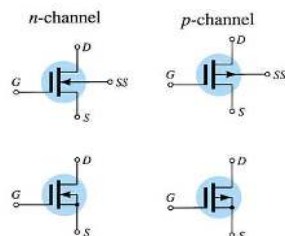
27

p-Channel D-Type MOSFET



28

D-Type MOSFET Symbols

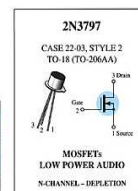


29

Specification Sheet

Maximum Ratings

Rating	Symbol	Value	Unit
Drain-Source Voltage	V_{DS}	20	Vdc
Gate-Source Voltage	V_{GS}	10	Vdc
Drain Current	I_D	20	mA
Total Device Dissipation @ $T_A = 25^\circ\text{C}$	P_D	200	mW
Derate above 25°C		1.14	mW/°C
Storage Temperature Range	T_{STG}	-55 to +200	°C



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30

Specification Sheet

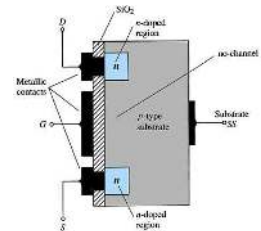
Electrical Characteristics

ELECTRICAL CHARACTERISTICS (T _{amb} = 25°C unless otherwise indicated)				
Characteristics				
OFF CHARACTERISTICS				
Static Drain-Source Leakage Voltage (V _{GS} = 0V, V _{DS} = 50V)	250/300	V _{DS(on)}	20	25
Gate-Source Capacitance (C _{gs}) (V _{GS} = 10V, V _{DS} = 0V) (V _{GS} = 10V, V _{DS} = 50V, f = 1MHz)	—	—	—	18
Drain-Source Capacitance (C _{ds}) (V _{GS} = 0V, V _{DS} = 50V, f = 1MHz)	250/300	V _{DS(on)}	—	—
Static Gate-Source Current (I _{gss}) (V _{GS} = 10V, V _{DS} = 0V)	—	I _{gss}	—	10
ON CHARACTERISTICS				
Zero-Channel Voltage On-Resistance (V _{GS} = 10V, V _{DS} = 0V)	250/300	R _{DS(on)}	2.0	2.0
On-State Drain Current (V _{GS} = 10V, V _{DS} = 50V)	250/300	I _{DS(on)}	9.0	14
SMALL-SIGNAL CHARACTERISTICS				
Transfer Characteristic (V _{GS} = 10V, V _{DS} = 0V, f = 1 kHz)	250/300	g _m	1000	1000
Output Admittance (V _{GS} = 10V, V _{DS} = 50V, f = 1 kHz)	250/300	h _{oe}	—	—
Input Capacitance (V _{GS} = 10V, V _{DS} = 0V, f = 1 kHz)	—	C _{iss}	—	18
Reverse Transfer Capacitance (V _{GS} = 10V, V _{DS} = 50V, f = 1 kHz)	—	C _{oss}	—	0.5
ESD PROTECTION CHARACTERISTICS				
Static Drain-Source Breakdown Voltage (V _{GS} = 0V, V _{DS} = 50V, I _{DS} = 100μA)	—	V _{DS(BR)}	—	50

31

E-Type MOSFET Construction

- The **Drain (D)** and **Source (S)** connect to the *n*-doped regions. These *n*-doped regions are connected via an *n*-channel
- The **Gate (G)** connects to the *p*-doped substrate via a thin insulating layer of SiO₂
- There is no channel
- The *n*-doped material lies on a *p*-doped substrate that may have an additional terminal connection called the **Substrate (SS)**

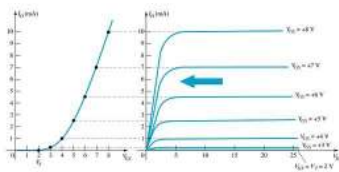


32

Basic Operation of the E-Type MOSFET

The enhancement-type MOSFET operates only in the enhancement mode.

- V_{GS} is always positive
- As V_{GS} increases, I_D increases
- As V_{GS} is kept constant and V_{DS} is increased, then I_D saturates (I_{DSS}) and the saturation level, V_{DSsat} is reached



33

E-Type MOSFET Transfer Curve

To determine I_D given V_{GS}:

$$I_D = k(V_{GS} - V_T)^2$$

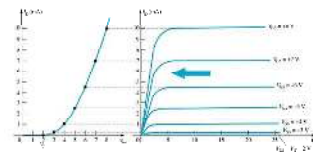
Where:
V_T = threshold voltage or voltage at which the MOSFET turns on

k, a constant, can be determined by using values at a specific point and the formula:

$$k = \frac{I_{D(ON)}}{(V_{GS(ON)} - V_T)^2}$$

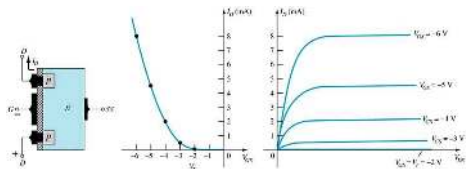
V_{DSsat} can be calculated by:

$$V_{DSat} = V_{GS} - V_T$$



34

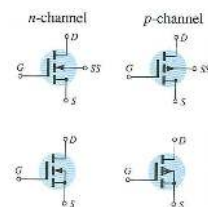
p-Channel E-Type MOSFETs



The *p*-channel enhancement-type MOSFET is similar to the *n*-channel, except that the voltage polarities and current directions are reversed.

35

MOSFET Symbols



36

Specification Sheet

Maximum Ratings

Symbol	Unit	Value	Unit
Power Dissipation	Watt	100	mW
Operating Voltage	Volt	0	50
Operating Current	mA	10	100
Operating Temperature	°C	0	50
Storage Temperature	°C	-55	+125
Lead Solder Temperature	°C	300	350
Lead Solder Time	sec	10	30

1. All dimensions are in millimeters unless otherwise specified.

2N4001
(CAT. NO. 100-101)
TO 100-102/103

Cathode
Anode
Mounting Lug

DIODE
SILICON
SILICON-GERMANIUM

more...

37

[illegible]

Handling MOSFETs

MOSFETs are very sensitive to static electricity. Because of the very thin SiO_2 layer between the external terminals and the layers of the device, any small electrical discharge can create an unwanted conduction.

Protection

- Always transport in a static sensitive bag
- Always wear a static strap when handling MOSFETs
- Apply voltage limiting devices between the gate and source, such as back-to-back Zeners to limit any transient voltage.

39

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CMOS Devices

CMOS (complementary MOSFET) uses a *p*-channel MOSFET and *n*-channel MOSFET; often on the same substrate as shown here.

The diagram illustrates a cross-section of a CMOS device. It features two MOSFETs on a common substrate. The left MOSFET is an n-channel MOSFET with a p-type substrate, a p-well, and an n+ source/drain region. The right MOSFET is a p-channel MOSFET with an n-type substrate, an n-well, and a p+ source/drain region. Both devices have a gate stack consisting of a gate oxide, a polysilicon gate, and a gate oxide. The gates are connected to a common gate line. The source and drain regions are connected to a common source/drain line. The diagram also shows the electrical connections for the gates, sources, and drains, labeled with 'G', 'S', and 'D' respectively.

Advantages

- Useful in logic circuit designs
- Higher input impedance
- Faster switching speeds
- Lower operating power levels

41

Summary Table

$I_{D1} = I_{D2} = I_{D3} = I_{D4}$ $I_{D1} = I_{D2} = I_{D3} = I_{D4} = I_{D10}$ $I_{D10} = I_{D10} \left(\frac{W_1/L_1}{W_{10}/L_{10}} \right)^2$	
$I_{D1} = I_{D2} = I_{D3} = I_{D4}$ $I_{D1} = I_{D2} = I_{D3} = I_{D4} = I_{D10}$ $I_{D10} = I_{D10} \left(\frac{W_1/L_1}{W_{10}/L_{10}} \right)^2$	
$I_{D1} = I_{D2} = I_{D3} = I_{D4}$ $I_{D1} = I_{D2} = I_{D3} = I_{D4} = I_{D10}$ $I_{D10} = I_{D10} \left(\frac{W_1/L_1}{W_{10}/L_{10}} \right)^2$	

42